forming molecules with lower vapor pressure than UF_6. Although research in these areas has indicated a large scale feasibility of several similar processes, no predictions can yet be made of their technological value. The energy consumption for the quantum processes discussed above are in the range 10–40 kWh SWU^{-1}. However, economic estimates indicate that their enrichment cost falls in the same price range as for centrifuge based plants. Hence, the interest in these methods has decreased and development of the ALVIS and SILVA processes has been terminated. Finally, it should be pointed out that the general concept of separative work is not valid for very high separation factors leading to very few units in a cascade (and always for mixtures of more than two isotopes).

3.9. EXERCISES

3.1. How many atoms of ^{235}U exist in 1 kg of uranium oxide, U_3O_8, made of natural uranium?

3.2. What is the atomic fraction of deuterium in water with the mole fraction of 0.81 for H_2O, 0.18 for HDO, and 0.01 for D_2O?

3.3. The translational energy of one mole of gas is given by $3/2\mathbf{R}T$, which corresponds to an average thermal molecular velocity v (the root mean square velocity), while the most probable velocity $v' = \sqrt{0.67}\, v$.

(a) What is the most probable velocity of a helium atom at 800°C?

(b) What voltage would be required to accelerate an α-particle to the same velocity?

3.4. A Nier type mass spectrometer has a fixed radius of curvature of 5 cm and a magnetic field of 3000 G; 1 G = 10^{-4} T. At what accelerating voltage will a Na^+ ion be brought to focus at the ion collector?

3.5. In a Dempster type (constant B and V) mass spectrograph utilizing 180° degree focusing, the ions $^{12}C^+$ and $^{11}BH^+$ are recorded simultaneously, the latter ion having a slightly larger orbit diameter. The separation between the lines recorded on the photographic plate is 0.0143 cm and the orbit diameter for the $^{12}C^+$ ion is 20 cm. What is the atomic mass of ^{11}B?

3.6. When the rotational quantum number n_r goes from 0 to 1 in $H^{35}Cl$, it is accompanied by the absorption of light with a wave number of 20.6 cm^{-1}. From this it is possible to calculate the interatomic distance between hydrogen and chlorine in the molecules. What is this distance?

3.7. In one mole of a gas at STP (standard temperature and pressure, i.e. 0°C and 1 atm) a small fraction of the molecules have a kinetic energy = $15\mathbf{k}T$. (a) How many such molecules are there, and (b) what would their temperature be if they could be isolated?

3.8. In an investigation the isotope ratio $^{18}O/^{16}O$ was found to be 2.045×10^{-3} for fresh water and 2.127×10^{-3} for carbon dioxide in the atmosphere. Calculate the equilibrium constant (mole fractions!) for the reaction

$$H_2{}^{18}O(l) + CO_2(g) = H_2O(l) + C\,{}^{18}O^{16}O(g)$$

3.9. How many ideal stages in an ordinary cascade are required at an $\alpha = 6$ to produce water in which 10% of the hydrogen is deuterium?

3.10. In a distillation column with total reflux, ^{10}B is enriched through exchange distillation of $BF_3O(C_2H_5)_2$ from the natural value of 20 atom % to a product containing 95% ^{10}B. The packed column has a length of 5 m and a diameter of 3 cm. What is the approximate height of a theoretical stage if the enrichment factor is 0.026?

3.11. A gas centrifuge plant is set up in order to enrich UF_6 of natural isotopic composition in ^{235}U. The centrifuges, which each have a length of 100 cm and a diameter of 20 cm, rotate at 40000 rpm. The gas temperature is 70°C.

(a) Prove that the separation factor α in (3.61) can be approximated by e^{δ} according to (3.75) when the product flow is very small compared to the waste flow, and α is not far from 1.

(b) Using this approximation, what is the theoretical separation factor for one unit?

(c) Assuming that the enrichment factor obtained with the centrifuge is only 70% of the theoretical one, what number of units would be required in series in order to achieve UF_6 with 3% ^{235}U?

3.12. How much separative work is needed in order to produce 1 kg of uranium containing 4% ^{235}U from a feed of natural uranium and leaving 0.25% in the tail?

3.10. LITERATURE

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